Reproducing EDM on CIFAR-10: Conditioning, Classifier-Free Guidance, and FID

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Abstract

Diffusion models generate images by learning to reverse a noising process, and the design space of this reversal is large. This report reproduces key results of Elucidating the Design Space of Diffusion-Based Generative Models (EDM, Karras et al., NeurIPS 2022) on CIFAR-10: unconditional and class-conditional sampling from NVIDIA's pretrained EDM networks, the behavior of the deterministic Heun sampler examined through denoiser trajectories, and classifier-free guidance (CFG), which mixes a conditional and an unconditional denoiser estimate with a weight w. We measure label accuracy, Fréchet Inception Distance (FID), and per-class FID across $w \in \{0, 1, 1.5, 2, 3, 4\}$ at 1,000 images per setting, and we train small CFG networks ourselves on CIFAR-10 subsets and on MNIST.

Code: https://github.com/vishudhshah/edm

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1 Introduction

Diffusion models generate images by progressively transforming random noise into structured samples through a learned denoising process (**?**). The denoising formulation behind modern samplers was unified by the score-based view (**?**), and EDM (**?**) re-derives the whole pipeline — noise schedule, network preconditioning, sampler, and training weights — as a design space of separable choices, reaching state-of-the-art FID on CIFAR-10 (1.97 unconditional, 1.79 conditional) with far fewer network evaluations than prior designs.

Producing realistic images does not by itself provide control over what those images depict. Classifier-free guidance (**?**) adds that control without a separate classifier: a conditional denoiser $D(x;\sigma,c)$ and an unconditional one $D(x;\sigma)$ are mixed as $D_w = D(x;\sigma)+w\,(D(x;\sigma,c)-D(x;\sigma))$, with $w > 1$ extrapolating along the direction the class makes in the estimate. Guidance improves prompt adherence but trades off sample variety, which motivates measuring accuracy, FID, and per-class FID together across w.

The checkpoint work here is the base for the capstone's goal of steering pretrained diffusion models with guidance-like signals. We reproduce: (1) unconditional and conditional sampling with EDM's pretrained CIFAR-10 networks, examined inside the sampler through trajectories of the noisy image and the denoiser's estimate; (2) the CFG sweep over w with label accuracy, FID, and per-class FID at 1,000 images per setting; (3) the training of small CFG networks on 100 and 1,000 CIFAR-10 images with 10% label dropout, including the memorization check against nearest training images; (4) the same CFG construction trained from scratch on MNIST; and, as compute allows, (5) a partial run of EDM's full 55.7M-parameter training.

1.1 Data

We use CIFAR-10 (**?**) (50,000 training and 10,000 test 32×32 color images in ten classes) for the pretrained-model experiments and the subset training, and MNIST (**?**) (60,000 28×28 grayscale digits) for the from-scratch CFG network. All FIDs use EDM's reference statistics of the 50,000 CIFAR-10 training images and EDM's Inception detector. With 1,000 samples per setting our FIDs are much higher than published 50,000-sample values, so runs are compared only with each other at equal sample counts.

2 Methods

3 Preliminary results

4 Discussion

5 Conclusion

References

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